

# Low-Ropes Force Map Relay

Predict balance before the balance challenge proves you right

**At a glance:** Predict balance and stability before the balance challenge proves you right. Plan about 80 minutes for the core block; 4 to 6 teams of 4 to 5 students.

## LEARNING OBJECTIVES

By the end of this activity, each student can:

- Explain the core science of this challenge: center of mass, balance, stability, force mapping (Understand).
- Apply the four fixed challenge cards safely and record a prediction before each test (Apply).
- Analyze whether the line of gravity stays over the base of support in each challenge (Analyze).
- Defend a center-of-mass claim with a prediction, observed result, and force map (Evaluate).

## MATERIALS FOR 25 STUDENTS AND 6 TEAMS

- balance challenge cards (this packet)
- clipboards
- center-of-mass templates (this packet), a backless chair, painter tape, and a coin and small weight
- pencils

## PREP BEFORE STUDENTS ARRIVE (15 TO 20 MIN)

- 01 Stage the four fixed challenge cards, a clear wall area, a backless chair with a helper spotter, a flat taped line, a coin, and a small weight.
- 02 Stage the scoring sheet and one center-of-mass template per team before rotations begin.
- 03 Clear the floor beside every challenge, assign a helper to spot the backless-chair stand, and keep the taped line flat. No running or raised balance surface.
- 04 Load the activity slides and ready the projected timer.

## MINUTE-BY-MINUTE FACILITATION

### Phase 1 Hook

0:00 - 0:05

Pose the mission as a challenge: "Predict balance and stability before the balance challenge proves you right." Take a quick prediction by show of hands.

## Phase 2 Prediction

0:05 - 0:12

Before each of the four fixed challenges, every team records whether it will work and where the center of mass will move.

## Phase 3 Constraints

0:12 - 0:18

Use only the four challenge cards. Test next to a clear wall, spot the chair stand, and keep the taped line flat and untimed.

## Phase 4 Run fixed challenges

0:18 - 0:44

Rotate teams through the four fixed challenge cards without revealing the center-of-mass explanation before each prediction.

## Phase 5 Test and record

Teams record the observed result and complete a force map for each challenge before moving to the next card.

## Phase 6 Evidence defense

Teams compare all four predictions with the observed results and defend one center-of-mass explanation.

## Phase 7 Score, debrief, cleanup

last 12 min

Teams submit a score slip, answer a debrief question with evidence, return the cards and templates, and clear the taped-line area.

## TROUBLESHOOTING

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**Problem:** A team invents a different pose. **Fix:** Return to the exact fixed challenge card so predictions and scoring remain comparable.

**Problem:** A challenge result is unclear. **Fix:** Recheck the card posture and base of support, then repeat once with the helper ready.

**Problem:** Teammates report different sensations. **Fix:** Score the observable result, then let each student mark where balance felt difficult.

**Problem:** A team finishes early. **Fix:** Ask them to improve one force map and explain the line of gravity without retesting a new pose.

## DIFFERENTIATION: THREE-TIER SUPPORT PROMPTS

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Tier 1 (stuck at the start):

- "What is the one science idea behind this challenge?" Point them to center of mass.

Tier 2 (completing challenges but not reasoning):

- "Where is the base of support, and where does the line of gravity fall?"

Tier 3 (ready to extend):

- "Compare two fixed challenges and explain why their bases of support produce different results."

#### DEBRIEF QUESTIONS (PICK 2 TO 3)

- When did your prediction match what you felt?
- How did lowering your center of mass help?
- Where else does balance physics matter?

#### SCORING RUBRIC (100 POINTS)

| SCORING CRITERION   | POINTS     |
|---------------------|------------|
| Correct predictions | 35         |
| Force-map quality   | 25         |
| Debrief explanation | 20         |
| Teamwork reflection | 20         |
| <b>Total</b>        | <b>100</b> |

#### CLEANUP AND SOURCES

Return the four challenge cards, templates, coin, and weight; move the chair out of the lane; remove tape if directed; and collect score slips.

**SOURCES** TeachEngineering center-of-mass and balance references